HW-1: Mapper Design for PAM-2 and 16-QAM

due: Apr. 1, 2026

1. Design a MATLAB program to realize a simple transceiver, as shown in Fig. 1, including a mapper, a channel, and a de-mapper for PAM-2 and 16-QAM modulation schemes, where the channel model is assumed as an ideal channel. A test bit streams ("**b.dat**") can be founded in E-Learning. The total number of bits is 100.

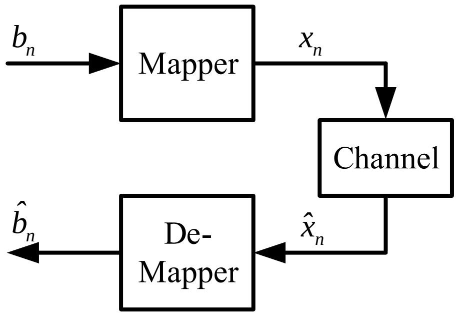

Fig. 1: Simple transceiver including a mapper, a channel, and a de-mapper.

The mapping rules of PAM-2 and 16-QAM are described as follows,

(1) For PAM-2, the mapping scheme is listed as

$$x_n = \begin{cases} +1, & b_n = 1 \\ -1, & b_n = 0 \end{cases}$$

where the number of symbols is equivalent to the number of information bits. Therefore, the total number of PAM-2 symbols is 100.

(2) For 16-QAM, the mapping scheme is shown as Fig. 2, for example, the 16-QAM symbol is equivalent to $-3 + j3$ when the information bits $\{b_n b_{n+1} b_{n+2} b_{n+3}\} = 0010$. The number of symbols is equivalent to $\frac{1}{4}$(the number of information bits), i.e., the total number of 16-QAM symbols is 25.

Problem:

Q1. Design a mapper and a de-mapper that can realize PAM-2 and 16-QAM. In other words, you can use a variable to do a selection for PAM-2 or 16-QAM. For example, **MTyp** = 0 for **PAM-2**, and **MTyp** = 1 for **16-QAM**, where **MTyp** can be viewed as **m**odulation **typ**e.

Q2. Based on "b.dat", you can design and evaluate a mapper for converting b_n to x_n.

Q3. Considering an ideal channel, $\hat{x}_n$ is equivalent to x_n. Therefore, you just need to generate x_n.

Q4. According to the results of x_n, you have to design a de-mapper for translating x_n to $\hat{b}_n$. It is no doubt that $\hat{b}_n$ is equivalent to b_n while your de-mapper is right.

Q5. Finally, you have to upload your **HW1_StudentID.m** file to E-Learning. The HW1_StudentID.m has two outputs (x_n and $\hat{b}_n$) for each modulation scheme, such as PAM-2 and 16-QAM. The x_n for PAM-2 and 16-QAM could be **PAM2_Symb** and **QAM16_Symb**, where the array sizes are 100 and 25, respectively. In other words, PAM2_Symb(1:100) and QAM16_Symb(1:25). Besides, The $\hat{b}_n$ for PAM-2 and 16-QAM could be PAM2_Bit and QAM16_Bit, where both array sizes are 100, namely, PAM2_Bit(1:100) and QAM16_Bit(1:100)

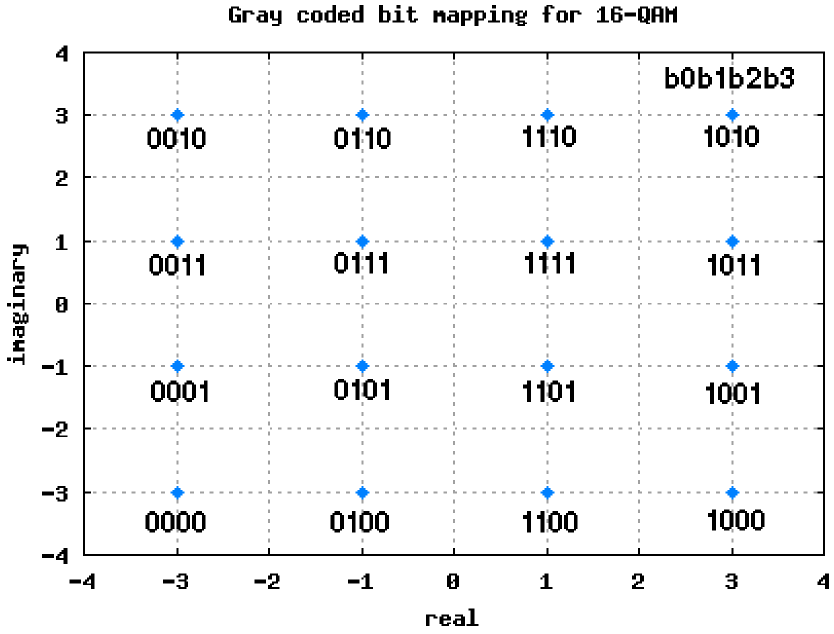

Fig. 2: 16-QAM Constellation